Chapter 3

Biot Savart's Law

The Biot-Savart law states that the magnetic field at a point from a small segment of current-carrying conductor is directly proportional to the current, the length of the element, and the sine of the angle between the element and the line joining it to the point. It is inversely proportional to the square of the distance between them.

This principle mathematically quantifies how current in conductors creates magnetic fields—central to all electromagnetism topics like magnetic effects of current and magnetism and electromagnetic induction

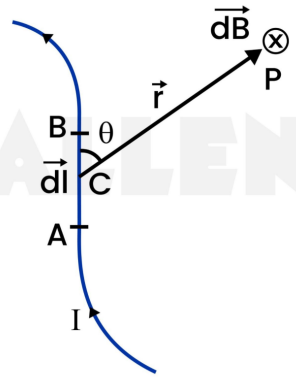
Statement of Biot Savart's Law

The Biot-Savart Law states that the magnetic field B at a point in space due to a small element of current-carrying conductor is directly proportional to the current I in the conductor, the length of the element dl, and the sine of the angle θ between the current element and the line connecting the element to the observation point where the field is being calculated. The magnetic field is inversely proportional to the square of the separation r from the element to the point.
Mathematically It can be expressed as,

$$dB = \frac{\mu_0}{4\pi}\frac{Idl\,\mathrm{Sin}\,\theta}{r^2}$$

Current Element-: A very small element of length dl of a thin current carrying conductor (wire). It is represented by $I(\overrightarrow{dl})$. It is a vector quantity and its direction is the same as that of current.

Diagram of Biot Savart's Law



Magnetic moment

Magnetic moment, also known as magnetic dipole moment, is the measure of the object's tendency to align with a magnetic field. The magnetic moment is a directional vector. A magnetic moment is a quantity that represents the magnetic strength and orientation of a magnet or any other object that produces a magnetic field. More precisely, a magnetic moment refers to a magnetic dipole moment, the component of the magnetic moment that can be represented by a magnetic dipole. A magnetic dipole is a magnetic north pole and a magnetic south pole separated by a small distance.

Susceptibility: The magnetic susceptibility (Latin: susceptibilis, “receptive”; denoted χ) is a measure of how much a material will become magnetized in an applied magnetic field. It is the ratio of magnetization M (magnetic moment per unit volume) to the applied magnetizing field intensity H.

$\chi = M/H$

Magnetism:

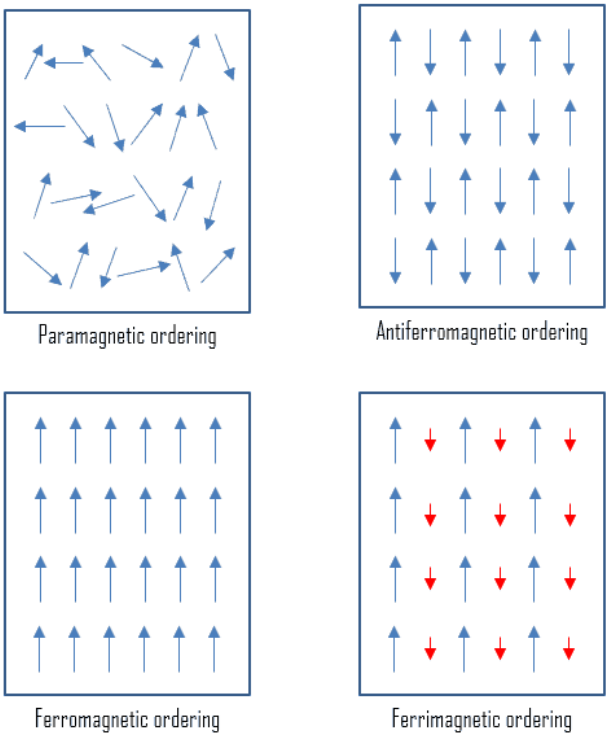
Magnetism is a phenomenon arising from moving electric charges, creating a magnetic field that exerts forces, leading to attraction or repulsion, on other magnetic objects and moving charges. A magnet is any material that can produce a magnetic field outside of itself.

Curie temperature

The Curie temperature is known as the temperature at which the ferromagnetic element starts to lose its magnetism and becomes paramagnetic when this temperature is exceeded. In the case that the material is below the Curie temperature, magnetic materials are transformed into ferromagnetic ones.

Curie’s law: It states that the susceptibility (the ability of a material to be magnetized by an external magnetic field) of a magnetic material is inversely proportional to the temperature.

Five basic types of magnetism have been observed and classified on the basis of the magnetic behavior of materials in response to magnetic fields at different temperatures. From this point of view, these types of materials are:



- Diamagnetic Material. Diamagnetic materials are repelled by a magnetic field; an applied magnetic field creates an induced magnetic field in them in the opposite direction, causing a repulsive force. Diamagnetism results from changes in electron orbital motion that are induced by an external field. Diamagnetic materials include water, wood, most organic compounds such as petroleum and some plastics, Diamagnetic and paramagnetic materials are considered nonmagnetic because the magnetizations are relatively small and persist only while an applied field is present. If χ (magnetic susceptibility) is negative, the material is diamagnetic. For example, the magnetic susceptibility of diamagnets such as water is $\chi_v = -9.05\times10^{-6}$. The most strongly diamagnetic material is bismuth, $\chi_v = -1.66\times10^{-4}$. Generally, nonmagnetic materials are said to be para- or diamagnetic because they do not possess permanent magnetization without external magnetic field.
- Bismuth: Bismuth is among the strongest diamagnetic materials and is often used in experiments to demonstrate diamagnetism.
- Water: Simple water is diamagnetic, which means that it can be levitated in sufficiently strong magnetic fields.

- Carbon Graphite: Graphite, an allotrope of carbon, shows significant diamagnetic properties, which are crucial in applications involving electromagnetic fields.
- Silicon: Widely used in the semiconductor industry, silicon is also known for its diamagnetic properties.

- Paramagnetic Materials: Paramagnetic materials are slightly attracted to magnetic fields. Paramagnetic materials are those having permanent atomic dipoles, which are acted on individually and aligned in the direction of an external field. Diamagnetic and paramagnetic materials are considered nonmagnetic because the magnetizations are relatively small and persist only while an applied field is present. If χ (magnetic susceptibility) is positive, a material can be paramagnetic. Paramagnetic materials include most chemical elements and some compounds. They have a relative magnetic permeability slightly greater than 1 (i.e., a small positive magnetic susceptibility) Generally, nonmagnetic materials are said to be para- or diamagnetic because they do not possess permanent magnetization without external magnetic field.
- Ferromagnetic Materials: Ferromagnetism is the basic mechanism by which a material form permanent magnet (i.e. materials that can be magnetized by an external magnetic field and remain magnetized after the external field is removed. Ferromagnetic, ferrimagnetic, or antiferromagnetic materials possess permanent magnetization even without external magnetic field. Every ferromagnetic material has its own individual temperature, called the Curie temperature, or Curie point, above which it loses its ferromagnetic properties. Only a few substances are ferromagnetic. The common ones are iron, cobalt, nickel and most of their alloys. Ferromagnetism is very important in industry and modern technology. Ferromagnetic materials can be divided into magnetically “soft” materials like annealed iron, which can be magnetized but do not tend to stay magnetized, and magnetically “hard” materials, which do.
- Antiferromagnetic Materials: In an antiferromagnet, unlike a ferromagnet, there is a tendency for the magnetic moments of neighboring valence electrons to point in

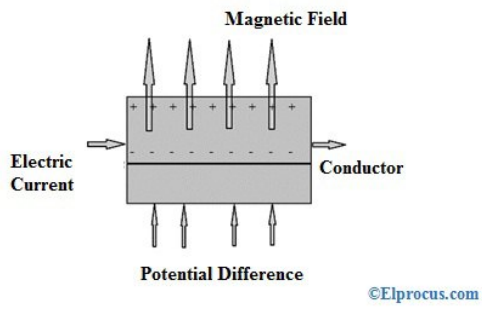
opposite directions. When all atoms are arranged in a substance so that each neighbor is anti-parallel, the substance is antiferromagnetic. Antiferromagnets have a zero net magnetic moment, meaning that no field is produced by them. Manganese oxide (MnO) is one material that displays this behavior.

- Ferrimagnetic Materials: A ferrimagnetic material is one that has populations of atoms with opposing magnetic moments, as in antiferromagnetism; however, in ferrimagnetic materials, the opposing moments are unequal and a spontaneous magnetization remains. Ferromagnetic, ferrimagnetic, or antiferromagnetic materials possess permanent magnetization even without external magnetic field . Magnetite (Fe_3O_4) is a famous example.

Properties	Ferromagnetic Materials	Paramagnetic Materials	Diamagnetic
State	They are solid.	They can be solid, liquid or gas.	They can be solid, liquid or gas.
Effect of Magnet	Strongly attracted by a magnet.	Weakly attracted by a magnet.	Weakly repelled by a magnet.
Behavior under non-uniform field	tend to move from low to high field region.	tend to move from low to high field region.	tend to move from high to low region.
Behavior under external field	They preserve the magnetic properties after the external field is removed.	They do not preserve the magnetic properties once the external field is removed.	They do not preserve the magnetic properties once the external field is removed.
Effect of Temperature	Above curie point, it becomes a paramagnetic.	With the rise of temperature, it becomes a diamagnetic.	No effect.
Permeability	Very high	Little greater than unity	Little less than unity
Susceptibility	Very high and positive	Little greater than unity and positive	Little less than unity and negative
Examples	Iron, Nickel, Cobalt	Lithium, Tantalum, Magnesium	Copper, Silver, Gold

Hall Effect:

Hall Effect was introduced by an American Physicist Edwin H.Hall in the year 1879. It is based on the measurement of the electromagnetic field. It is also named as ordinary Hall Effect. When a current-carrying conductor is perpendicular to a magnetic field, a voltage generated is measured at right angles to the current path. Hall Effect is defined as the difference in voltage generated across a current-carrying conductor, is transverse to an electrical current in the conductor and an applied magnetic field perpendicular to the current.

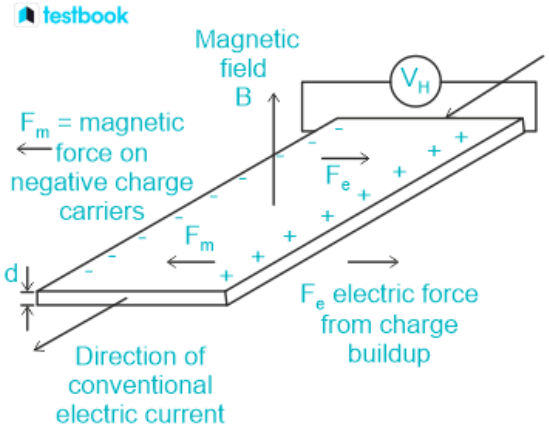


When an electric current flows through a conductor, electrons move along its length in the opposite direction of the current. When a magnetic field is applied perpendicular to this current, something interesting happens. Each electron feels a force called the Lorentz force, which pushes it sideways, perpendicular to both the current direction and the magnetic field lines, making them move to one side of the conductor. As electrons gather on one side, they create a negative charge buildup, leaving a positive charge on the other side. This separation of charges creates an electric field inside the conductor that opposes further electron movement.

This process continues until the electric force from the charge buildup balances the magnetic force from the Lorentz force. The result is a stable situation where the forces are balanced, and a voltage, called the Hall voltage, is generated across the conductor.

Hall Effect

When the current flows through the conductor under the presence of the magnetic field, the charge carriers which are the electrons experience a transverse force which pushes them to one side of the conductor. This causes an excess positive charge on the other side of the conductor.



Due to this buildup of charge carriers on one side of the conductor a potential difference gets established between the sides of the conductor. The existence of this transverse voltage which is measurable is known as the Hall Effect.

If current I is flowing in the conductor, then it can be defined as $I = nev_dA$, where n is the number of electrons per unit volume, e is the charge on electrons, v_d is a small unit of volume and A is the area of cross-section of the conductor.

The equations of v_d and **I** can be combined to get the following equation :

$I = ne(\frac{E}{B})A$ where B is the magnetic flux density in Tesla and E is the electric field.

The electric field **E** is related to the potential difference **V** by :

$E = \frac{V}{l}$ where **V** is the potential difference across the conductor and **l** is the length of the conductor across which the voltage is measured.

Combining the equations for ***E*** and ***I*** we get,

$V = \frac{IBL}{neA}$

Where **V** is the Hall Voltage developed across the length of the conductor.

Hall Effect can also be derived in a different way.

If a magnetic field B is applied in the positive Z direction for a conductor carrying current in the positive Y direction then Lorentz Force on the electrons is give by :

$F_l = -Bev_d$ where e is the magnitude of charge and v_d is the drift velocity.

The Lorentz force is exerted in negative Y direction and thus due to the Hall Effect there is a charge build up on one side which causes positive charge buildup on the other side too.

Due to this an electric field gets generated which is given by :

$F_e = -eE_h$ where F_e is the force due to the electric field and E_h is the electric field generated due to the hall effect.

Since Lorentz force and electric force are balanced in a neutral situation thus,

Since Lorentz force and electric force are balanced in a neutral situation thus,

$F_e = F_l$

Thus we get $E_h = Bv_d$

If b is the width of the conductor then $E_h = \frac{V_h}{b}$ or $V_h = E_h b$ which can be stated as $V_h = Bv_d b$ 1

Now, if J is the current density it can be stated as $J = -n_e e v_d$.

Substituting the above equation in Equation 1 we get,

$V_h = -Bb\frac{J}{n_e e}$

But $V_h = E_h b$ we get,

$E_h = \frac{BJ}{n_e e}$ which is the formula for the electric field generated due to the Hall Effect.

Hall Coefficient

Hall coefficient is stated as R_H which is given by $R_H = \frac{1}{ne}$ where n is the number of electrons.

Hall Coefficient formula for semiconductors become negative when the number of electrons are more than the holes.

Lorentz force: The Lorentz Force is the combined effect of electric and magnetic forces acting on a charged particle. When a charged particle moves through both an electric field and a magnetic field, it experiences a force that can change its trajectory. The force is perpendicular to both the particle's velocity and the magnetic field direction.

The Lorentz Force is a fundamental concept in electromagnetism that describes the force experienced by a charged particle moving through an electric and magnetic field. This force, named after the Dutch physicist Hendrik Lorentz, plays a crucial role in understanding the behavior of charged particles in various physical contexts. Mathematically, it is expressed as F = q(E + v × B), where F is the force, q is the charge, E is the electric field, v is the velocity, and B is the magnetic field. In real life, the Lorentz Force is observed in numerous applications, such as in the operation of electric motors and generators, where it helps convert electrical energy into mechanical energy and vice versa. It's also fundamental to the working of devices like cyclotrons and mass spectrometers, which are used in medical treatments and scientific research. Even in our daily lives, the principles behind the Lorentz Force can be seen in the functioning of devices like smartphones and televisions, making it an essential concept in both advanced technology and everyday gadgets. In this article, we will discuss the concept of Lorentz force and solve examples for concept clarity.

Ampers law:

Ampere's Law states that the line integral of the magnetic field around any closed loop is directly proportional to the total electric current passing through the loop. In simple words, the magnetic field produced in space is always closely linked with the movement of electric charges.

Mathematically, Ampere's Law is expressed as:

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc}$$

Here,

- **B**: Magnetic field vector
- **dl**: Small element along the closed path (loop)
- μ_0: Permeability of free space ($\mu_0 = 4\pi \times 10^{-7}$ T·m/A)
- I_{enc}: Net current enclosed by the path

The line integral of the magnetic field B around an arbitrarily chosen path l is proportional to the net electric current I enclosed by the path. That is, $B \cdot dl = \mu_0 I$,where μ_0 is the magnetic permeability of free space and is $4\pi \times 10^{-7}$ weber per ampere-metre.

Difference between Biot-Savart law and amperes law:

1) Biot-Savart law is the basic equation of magnetic field due to the current distribution.
2) When electric field is provided by the gradient of potential and electric flux changes in time then the Biot-Savart law is valid.
3) Biot-Savart law is follows the concept of magnetism.
4) Asymmetric current distribution is valid for Biot-Savart law.
5) Biot-Savart law is the differential form of magnetic field induction B.
6) It is quite difficult to find the magnetic field by the integrals of Biot-Savart law.
7) The integral form of Biot-Savart law is : $B\ dl = \mu_0 I$

1) Ampere's law gives the another method to calculate magnetic field of given current distribution.
2) When the electric field through the loop is constant in time then Ampere's law

is valid.

3) Ampere's law follows the concept of of electromagnetism.

4) Symmetric current distribution is valid for Ampere's law.

5) Ampere's law is the integral form of magnetic field induction B

6) Finding magnetic field by Ampere's law is much simpler than the Biot-Savart law

7) The differential form of Biot-Savart law is : $B = \mu_0 NI/2R$